

TdhGIS API

Overview
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Introduction

[Python bindings](#) are now provided for the TdhGIS API.

Spatial analysis packages are unwieldy and difficult to use, until now. How about a package that:

- Can perform spatial analysis using just 2 executable files (less than 3 Mb, uncompressed) and a handful of c++ include files. No other external libraries required. This is a lean and mean calculating machine.
- Can calculate:
 - polygon area, perimeter and centroid
 - thiessen polygons from points
 - contour lines and polygons from points
 - polygon intersection, merge, subtraction, split and trim
 - whether a point or polygon lies within a polygon and the point on a polygon closest to a specified point
- Can use geometry data from existing code, no data duplication necessary.
- Uses linked list data structures so there are no built in limits on the number of any components. The package will scale nicely from modest to high end hardware.
- Interfaces with a library to perform database I/O, requiring just sqlite.
- Interfaces with the GDAL package for data transfer with a wide variety of formats, while optionally performing coordinate transformation.
- Includes libraries for linux (gcc), msvc and mingw.

These features are highlighted in compact demonstration source code that will get you going in the time it takes to download other packages.

Libraries and Header Files

The header files contain documentation for the included classes and functions.

The library files are contained in directories associated with one of 3 compilers:

- gcc for linux
- msvc for mswin
- mingw for mswin

2 libraries are needed to perform spatial analysis:

- TdhCommon provides general data structure management, defined in the file TRecordsNav0.h, and geometry specific data structures, defined in the file CadPrimitives0.h. Some basic spatial analysis functionality not specific to polygons is also provided here.
- TdhSpatial performs the heavy lifting for spatial analysis with the interface defined in TdhSpatial_Intf.h

These libraries and header files are the same as used for the TdhGIS app, so they can support everything from a simple to a robust spatial analysis project.

The TdhGIS_API library is used to perform database I/O requiring only standard sqlite. The interface is provided in the TdhGIS_API.h header file. The database structure is the same used for the TdhGIS app, so the app can be used to create input data and inspect results.

The TdhOGR_API library makes use of the gdal package to exchange data with a wide variety of formats with the TdhGIS_API package. The interface is provided in the TdhOGR_API.h header file. This library facilitates coordinate transformation during a data exchange process. The capabilities are essentially the same as provided in the TdhOGR app. The libraries TdhCairo and TdhPath are also needed because the TdhOGR_API library also data exchange with TdhCad and TdhGISnet, in addition to Open Street Map.

Along with the power of gdal comes the challenges of using such a large and multifaceted package. The gdal package is not distributed with TdhGIS_API. The gdal package used to compile and link TdhOGR_API was obtained through standard channels (e.g. pkg-config, vcpkg and pacman, for gcc, msvc and mingw, respectively). Successful use of TdhOGR_API is more likely with gdal packages issued about the same time as the TdhGIS API package.

Demonstration App

The demonstration app, contained in the demo_source directory of the distribution package, is intended show how to use the TdhGIS API quickly and easily. It is intended to be run as a console app.

The directory has a source/header file combination for each of the 3 library groups described above and a main.cpp file that calls each of the 3 other source files. The calls to the other 3 source files can be implemented independently and the main.cpp starts off with only the call to TdhSpatial_API, the most fundamental demonstration, not commented out. After each call to a source file is successfully executed and understood, it can be commented out and the next one uncommented.

To make use of the last 2 source source files, which demonstrate the TdhGIS_API and TdhOGR_API libraries, the variable dataDir in main.cpp must be set to the location of the Data directory that was included with distribution package.

As mentioned previously, the gdal package is not included in the distribution package for the TdhGIS API, so, to make use of the TdhOGR_API_demo source file, the gdal package will need to be obtained for the compiler in use.

Python Bindings

TdhGIS_API now provides python bindings so that virtually all the functionality provided by the API is accessible through python. The notable exception is the ability to access x,y data directly from user defined python data structures, as can be done with C++. When using python, x,y data must be read or copied into TdhGIS data structures.

Python must have access to the same libraries provided for C++, plus the libraries that provide the python bindings. All such libraries are provided in the bin directory of the API download file. Otherwise, nothing beyond standard python is required.

A series of python classes corresponding to the C++ API classes are provided in the python directory of the API download file. Documentation is provided with each function definition.

Python versions of the demonstration code are provided in the python/TdhGIS_API_demo_py directory. The demonstration code can be executed directly from a command prompt by using the script provided in the python directory.

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